

Question: Create a PHP function that finds all prime numbers in a given range and displays them.

Algorithm Overview

The Sieve of Eratosthenes works by iteratively marking the multiples of each prime number starting from 2. The multiples of a given prime are generated as a sequence of numbers starting from that prime, with a constant difference between them equal to that prime. This is key to the algorithm's efficiency.

Implementation

We'll define a function findPrimeNumsInRange that takes a start and end range and returns all prime numbers within that range. This function will use the Sieve of Eratosthenes algorithm for efficiency.

```
<?php

function findPrimeNumsInRange($start, $end) {
    // Initialize a boolean array "prime[0..n]" and initialize all entries it as true.
    // A value in prime[i] will finally be false if i is Not a prime, otherwise true boolVal.
    $prime = array_fill(2, $end + 1, true);

    // Iterate from 2 to square root of n
    for ($p = 2; $p*$p <= $end; $p++) {
        // If prime[p] is not changed, then it is a prime
        if ($prime[$p] == true) {
            // Update all multiples of p
            for ($i = $p*$p; $i <= $end; $i += $p) {
                $prime[$i] = false;
            }
        }
    }

    // Collect and return all prime numbers in the specified range
    $primeNumsInRange = array_filter(array_keys($prime), function($num) use ($start, $end) {
        return $num >= $start && $num <= $end;
    });

    return $primeNumsInRange;
}

// Example usage
$start = 10;
$end = 50;
$primeNumbersInRange = findPrimeNumsInRange($start, $end);
echo implode(' ', $primeNumbersInRange);
```

Explanation

Initialization: We start by initializing a boolean array \$prime with all values set to true. This array represents whether each index is a prime number.

Marking Multiples: We iterate through each number from 2 to the square root of the end of the range. If a number is still marked as prime, we mark all its multiples as not prime (false).

Filtering Primes: After processing, we filter the \$prime array to collect only the prime numbers within the specified range.

Displaying Results: Finally, we join the prime numbers into a string and echo it out.